

Investigating the Behavioural Sensitivity of Aircraft Boarding Strategies: An Agent-Based Simulation Study

Sanyal, Srinjoy; Vanhoonacker, Louis; Yu, Tao

Abstract

Aircraft boarding efficiency affects aircraft turnaround and operational performance, yet many boarding studies assume ideal passenger behaviour and overlook behavioural variability. This study evaluates the performance and behavioural sensitivity of five aircraft boarding strategies using an agent-based simulation model. More realistic passenger behaviour was represented through passenger social-group formation, allowing travellers to partially coordinate boarding behaviour. Boarding performance was assessed using total boarding time together with selected behavioural indicators. Results show that under ideal conditions, WilMA and the Steffen Method achieved the shortest boarding times, while Front-to-Back performed worst. Introducing passenger groups did not substantially alter the ranking of strategies but affected their robustness differently. Strategies relying on stricter boarding order showed greater sensitivity to behavioural variation, whereas WilMA remained comparatively stable across scenarios. These findings suggest that evaluating boarding methods under realistic behavioural assumptions is important and that strategy robustness may be as relevant as theoretical efficiency.

Keywords: aircraft boarding, agent-based modelling, behavioural sensitivity, passenger behaviour

1. Introduction

Aircraft boarding is an important operational process that affects airline efficiency, passenger experience, and aircraft turnaround time. Although boarding represents only one stage of airport operations, delays during this process can propagate throughout flight schedules and contribute to increased operational costs. Consequently, improving boarding efficiency has attracted sustained interest from both industry and academic research.

Previous studies have evaluated boarding strategies such as Random Boarding, Back-to-Front (B2F), and Window-Middle-Aisle (WilMA), often under idealised assumptions in which passengers strictly follow boarding instructions and act independently. However, real passenger behaviour is more variable. Travellers may board with family members or companions and coordinate their movement rather than strictly follow the assigned sequence. Such behaviour can partially disrupt boarding order and alter passenger flow. From a complex systems perspective, aircraft boarding can therefore be viewed as a Complex Adaptive System (CAS), where overall performance emerges from interactions among passenger behaviour, boarding rules, luggage handling, seat interference, and limited behavioural coordination.

This study investigates how different boarding strategies perform under both ideal and more realistic behavioural conditions. More specifically, it examines whether introducing social-group behaviour changes boarding efficiency and affects the robustness of different boarding methods. Boarding

strategies are evaluated using operational indicators including total boarding time, passenger standing time, and seat shuffle delay.

2. Model Design and Experimental Setup

2.1 Model Overview

An agent-based simulation model was developed in AnyLogic to evaluate aircraft boarding efficiency under different behavioural conditions. The model represents a single-aisle aircraft with 28 rows and 6 seats per row (168 passengers), including the boarding gate, queueing area, aircraft aisle, and assigned seating layout. Each agent represents an individual passenger and is assigned seat location, hand-luggage characteristics, and behavioural attributes that influence movement and boarding interactions.

2.2 Passenger Behaviour

Passenger behaviour is implemented using a statechart in AnyLogic (Figure 1), which represents the boarding process of each passenger agent from entering the queue to reaching the assigned seat.

Passengers begin in the WaitingAtGate state and enter boarding once either their assigned boarding group or a member of their social group is called. Boarding order is determined by seat location and the selected strategy, while social-group membership may cause passengers to board together. Passengers then progress through Queuing and Walking, where movement is constrained by available space and may generate temporary congestion. Upon reaching their assigned row, passengers store carry-on luggage (Stowing), searching for alternative overhead space if necessary (FoundEmptyBin). Finally, passengers move to their seats (MovingToSeat), where seat interference may introduce delays (SeatShuffleDelay), before reaching the terminal Seated state.

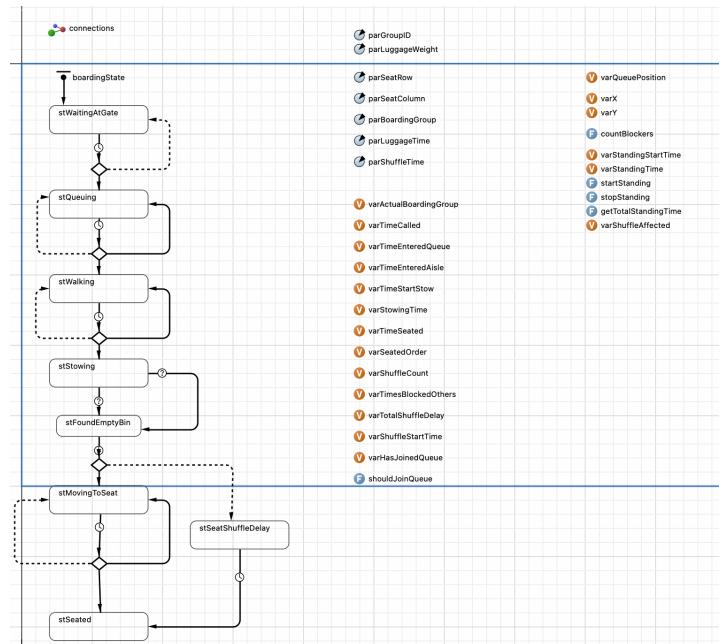


Figure 1. Simplified passenger statechart illustrating boarding, movement, luggage stowing, and seating processes.

2.3 Model Assumptions and Parameterisation

Several model parameters were informed by findings from previous studies and industry reports. Passenger baggage characteristics were parameterised using data from the European Aviation Safety

Agency (EASA, 2009; EASA, 2022). In the model, hand-luggage weight was generated using a normal distribution with a mean of 6.1 kg and a standard deviation of 4.6 kg to reflect variation across passengers. To represent luggage-related constraints during boarding, overhead storage capacity was limited to 44 kg per compartment based on recommendations reported by the Australian Transport Safety Bureau (ATSB). This capacity constraint influences luggage storage behaviour by restricting the available capacity within nearby overhead bins. Luggage stowing behaviour was modelled as a stochastic process rather than a fixed delay. This approach is supported by experimental findings demonstrating substantial variation in luggage storage times between passengers (Ren et al., 2020).

Passenger social behaviour was incorporated to represent the tendency of travellers to board together with accompanying passengers. Social-group formation was included to capture realistic boarding behaviour and was informed by previous studies reporting that approximately 10–20% of passengers travel in groups, with pairs representing the most common group structure (Bachmat et al., 2023).

2.4 Experimental Design

To investigate the behavioural sensitivity of aircraft boarding strategies, a two-factor experimental design was adopted, consisting of Boarding Strategy and Social-Group Behaviour Scenario. Five boarding strategies were evaluated: Random Boarding (no predefined order), Back-to-Front (B2F; rear-to-front sequence), Front-to-Back (F2B; front-to-rear sequence and included as a contrast condition), Window-Middle-Aisle (WilMA; boarding by seat position), and the Steffen Method (alternating rows and seat positions to reduce interference). These strategies represent different levels of boarding structure and are commonly used in aircraft boarding research (Schultz, 2017; Delcea et al., 2023).

Social-group behaviour was controlled using two model parameters: the probability of travelling in groups (parGroupChance) and the probability of forming groups larger than two passengers (parLargeGroupChance). Three scenarios were considered: ideal (0.0, 0.0), representing fully independent boarding; realistic (0.25, 0.20), representing occasional small-group travel; and extreme (0.40, 0.40), representing frequent and larger group formation.

For each combination of boarding strategy and behavioural scenario, 10 simulation runs with different random seeds were conducted. Each run simulated the complete boarding of 168 passengers. Performance was evaluated using Total Boarding Time (TBT) as the primary outcome, defined as the time from boarding initiation until the final passenger was seated. Additional indicators included standing time, luggage stowing time, seat shuffle events, and passenger blockage measures. Results were exported for statistical comparison across scenarios.

3. Results

3.1 Boarding Strategy Comparison under Ideal Social-Group Conditions

Strategy	Runs	Boarding Time (Mean $\pm$ SD)	Standing Time (Mean $\pm$ SD)	Total Shuffle Delay (Mean $\pm$ SD)
WMA	10	1511.39 $\pm$ 53.25	340.29 $\pm$ 11.34	0.00 $\pm$ 0.00

Steffen	10	1515.32 ± 74.66	241.31 ± 12.71	0.00 ± 0.00
Random	10	1781.12 ± 73.91	292.86 ± 14.99	1911.78 ± 201.47
B2F	10	2358.55 ± 160.76	451.79 ± 28.79	1785.49 ± 240.32
F2B	10	2920.69 ± 146.04	490.65 ± 22.94	1863.39 ± 195.10

Table 1 Summary statistics of boarding performance under ideal social-group conditions (parGroupChance = 0.0, parLargeGroupChance = 0.0)

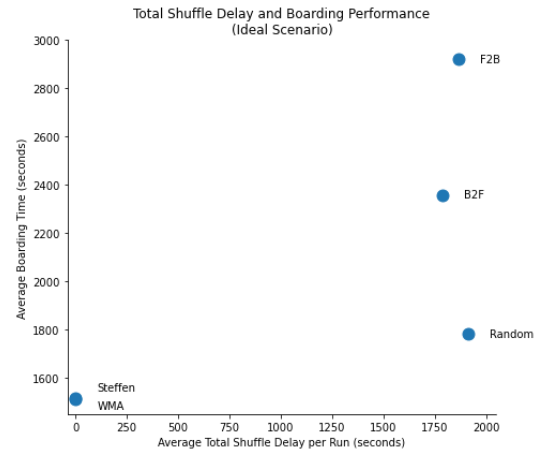
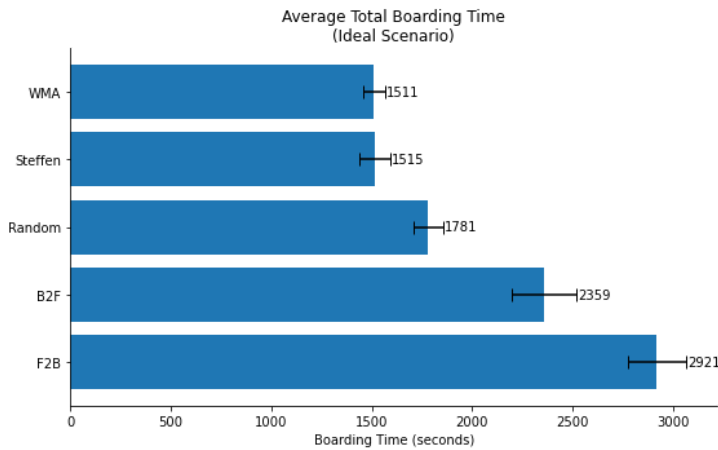


Figure 2. Average boarding time across strategies under ideal conditions (mean ± SD, n = 10). Figure 3. Relationship between average shuffle delay and boarding time under ideal conditions. *WilMA and Steffen overlap due to similar values.*

Under ideal conditions (parGroupChance = 0.0, parLargeGroupChance = 0.0), clear performance differences were observed across boarding strategies (Table 1; Figure 2). WilMA achieved the shortest average boarding time (1511.39 ± 53.25 s), closely followed by the Steffen method (1515.32 ± 74.66 s). Although the difference between these two strategies was small, both substantially outperformed Random Boarding (1781.12 ± 73.91 s) and the row-based approaches. Back-to-Front required considerably longer boarding time (2358.55 ± 160.76 s), while Front-to-Back produced the poorest performance overall (2920.69 ± 146.04 s). Additional behavioural indicators revealed differences that were not reflected in boarding time alone. Despite achieving similar total boarding durations, WilMA and Steffen showed distinct interaction patterns. Steffen produced lower average standing time (241.31 s) than WilMA (340.29 s), indicating reduced waiting time during the boarding process. At the same time, both strategies recorded zero shuffle delay under ideal conditions, in contrast to Random Boarding (1911.78 s), Back-to-Front (1785.49 s), and Front-to-Back (1863.39 s).

Figure 3 further highlights that shuffle delay alone did not fully determine boarding efficiency. Although Random Boarding generated substantial shuffle delay, it still outperformed both row-based strategies. Similarly, Back-to-Front and Front-to-Back produced comparable average shuffle delay values but differed considerably in total boarding time. These observations suggest that seat interference was not the sole source of delay and that boarding efficiency was also influenced by passenger waiting and the concentration of passenger movement within the aisle.

3.2 Behavioural Sensitivity under Social-Group Conditions

Strategy	Group/Large	Runs	Avg TBT	SD	Change (%)
B2F	0 / 0	10	2359	161	0.0
	0.25 / 0.20	10	2450	70	+3.9
	0.40 / 0.40	10	2386	141	+1.2
F2B	0 / 0	10	2921	146	0.0
	0.25 / 0.20	10	2863	152	-2.0
	0.40 / 0.40	10	2893	159	-0.9
Random	0 / 0	10	1781	74	0.0
	0.25 / 0.20	10	1790	96	+0.5
	0.40 / 0.40	10	1834	97	+3.0
Steffen	0 / 0	10	1515	75	0.0
	0.25 / 0.20	10	1574	83	+3.8
	0.40 / 0.40	10	1616	98	+6.6
WMA	0 / 0	10	1511	53	0.0
	0.25 / 0.20	10	1567	83	+3.6
	0.40 / 0.40	10	1555	51	+2.9

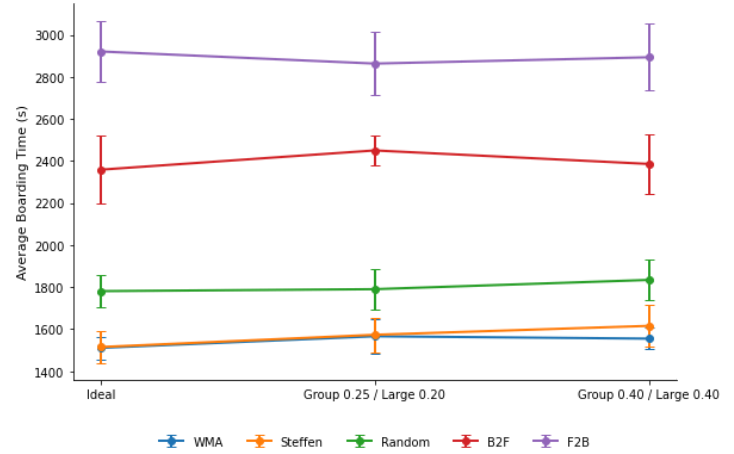


Figure 4. Sensitivity of boarding performance under different social-group scenarios; Figure 5. Average boarding time across social-group scenarios for each boarding strategy. Error bars represent standard deviation across simulation runs.

To evaluate whether these performance patterns remained stable under more realistic passenger behaviour, social-group formation was introduced through three behavioural scenarios: ideal (0.0, 0.0), realistic (0.25, 0.20), and extreme (0.40, 0.40). Overall, social-group formation produced only moderate changes in boarding time and did not substantially alter the ranking of strategies. However, the magnitude of performance change varied across methods. WilMA and Steffen remained the two fastest approaches under all scenarios. Under ideal conditions, WilMA achieved the shortest average boarding time (1511.39 ± 53.25 s), closely followed by Steffen (1515.32 ± 74.66 s). As social-group behaviour increased, Steffen experienced the largest deterioration, rising to 1573.60 s and 1616.03 s under the moderate and extreme scenarios respectively (+6.6% relative to the ideal condition). By comparison, WilMA showed only limited increases and maintained the best overall performance. Random Boarding also remained relatively stable across scenarios, with only minor increases in boarding time. In contrast, the row-based strategies showed weaker and less systematic responses: Back-to-Front exhibited moderate fluctuations without a consistent increasing trend, whereas Front-to-Back remained consistently inefficient across all conditions.

Figure 5 further illustrates that increasing social-group formation mainly affected strategy robustness rather than overall effectiveness. Strategies relying on stricter passenger ordering, particularly Steffen, became more sensitive to behavioural variation, whereas WilMA and Random Boarding maintained comparatively stable performance. This pattern suggests that boarding methods requiring stronger adherence to predefined passenger sequences may become more vulnerable when passengers coordinate their movement and partially deviate from the intended boarding order.

4. Discussion and Conclusion

This study investigated how aircraft boarding strategies perform under both ideal and more realistic behavioural conditions using an agent-based simulation model developed in AnyLogic. Passenger

behavioural variation was represented through social-group formation, allowing evaluation of whether boarding efficiency remains stable when passengers do not act fully independently. Results showed that introducing passenger groups did not substantially alter the overall ranking of boarding strategies but affected their robustness differently. Strategies relying more strongly on strict boarding order became increasingly sensitive to behavioural variation, whereas WilMA remained comparatively stable across scenarios. These findings suggest that evaluating boarding methods solely under ideal operational assumptions may overestimate the practical advantages of highly structured boarding procedures. Under the assumptions of the present model, WilMA provided the most favourable balance between boarding efficiency and behavioural robustness.

Several limitations should be acknowledged. Passenger coordination was represented through simplified social-group behaviour and did not capture richer interaction mechanisms or adaptive decision-making during boarding. In addition, overhead luggage competition was not modelled as a binding constraint because the selected storage-capacity assumptions provided sufficient space for all passengers. Future work could incorporate more realistic behavioural rules, dynamic passenger interaction, and constrained storage conditions to further examine how operational complexity influences boarding performance.

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